

Slide	Notes
C O D E Lesson 5: Uncovering Contradictions Problem Solving with AI	
Warm Up	Warm Up (10 mins)
Problem Solving with AI Uncovering Contradictions – Warm Up ✔ Do This: Choose a topic! For example, space, animals, history, music, etc. 	Two Truths and a Lie (10 mins) ■ ■ Do This: Ask the class for a topic (e.g., space, animals, history, music, etc.).
Problem Solving with AI Uncovering Contradictions – Warm Up Two Truths and a Lie Lesson 5 1 Let's see what the AI gives us when I enter the following prompt: Tell me 2 truths and a lie about [topic]. Don't tell me which ones are truths or lies. 	■ ■ Do This: Use Lesson 5, Level 1 to demo entering the following prompt: "Tell me 2 truths and a lie about [insert topic]. Don't tell me which ones are truths or lies."
Problem Solving with AI Uncovering Contradictions – Warm Up ✔ Do This: With a partner, try to guess which AI-generated statements are true and which is the lie. 	■ ■ Do This: Have students partner up and try to guess which AI-generated statements are true and which is the lie.

Problem Solving with AI Uncovering Contradictions – Warm Up Fact-Checking Let's see how we can verify the information the AI gave us. DEMO 	■ Say: AI might not always produce the results that is expected. The result might even be contradictory. Therefore, fact-checking AI's results is an important step of the process. ■ Do This: Demonstrate fact-checking by researching online using Google and credible websites to verify which statements are true and which is a lie.
Problem Solving with AI Uncovering Contradictions – Warm Up  Discuss: Why might AI generate a lie or inaccurate information? How do we ensure accuracy when using AI? 	■ Discuss: Click through the animated slide to display the prompts. Why might AI generate a lie or inaccurate information? How do we ensure accuracy when using AI? ■ Discussion Goal: Students should understand that AI generates responses based on patterns in data rather than verified facts. They should discuss how AI lacks direct knowledge, can misinterpret ambiguous prompts, and sometimes "hallucinates" information because it is designed to predict plausible-sounding text rather than confirm truth. Student highlight the importance of human judgement in verifying AI-generated content. Students should identify fact-checking strategies, such as cross-referencing multiple reliable sources (using official or expert-verified information) and critically analyze responses for logical consistency.
Problem Solving with AI Uncovering Contradictions – Warm Up Lesson Objectives By the end of this lesson, you will be able to . . . • Experiment with AI to uncover contradictions and explain why they occur. • Analyze how AI's reliance on probability-based models can lead to inconsistencies. • Evaluate the implications of AI-generated contradictions in real-world applications.	■ Do This: Introduce the lesson objectives.
Problem Solving with AI Uncovering Contradictions – Warm Up Question of the Day Can AI contradict itself?	■ Do This: Introduce the Question of the Day.
Activity	Activity (30 mins)

What is a Contradiction?

A **contradiction** is when an AI generates conflicting or opposing responses based on a similar input, making it impossible for both responses to be correct at the same time.

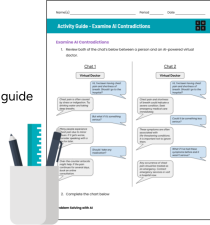
**Contradictions in Real World Contexts (15 mins)**

■ Do This: Define contradiction.

Contradictions in Real World Contexts

You and your partner should have:

- Examining AI Contradictions activity guide
- pen or pencil



■ Distribute: Give each pair of students a copy of the Examining AI Contradictions activity guide.

Contradictions in Real World Contexts**Do This:**

1. In your activity guide, review both AI-generated responses.
2. Based on your observations, complete the AI Contradiction Chart.
3. Fact check AI's claims using reliable sources.
4. Refine prompt to get more a consistent response.

■ Do This: Using their activity guide, students will review two contradictory AI responses from a telehealth chatbot.

Step 1: Students review both AI-generated responses.

Step 2: Students highlight contradictions, vague statements, or misleading claims, such as the conflicting advice on whether or not immediate medical attention is necessary.

Step 3: Students fact-check the AI's claims using reliable sources (e.g., trusted medical websites or professional guidelines).

Step 4: Students refine the provided prompt to get a more consistent and clearer response from the AI.

Discuss:

How does the AI's lack of specialization in the medical field contribute to contradictions?

How can fact-checking and prompt refinement help improve AI-generated responses in real-world contexts?



■ Discuss: Click through the animated slide to display the prompts.

How does the AI's lack of specialization in the medical field contribute to contradictions?

How can fact-checking and prompt refinement help improve AI-generated responses in real-world contexts?

■ Discussion Goal: Students should recognized that AI models do not have deep expertise like medical professionals. They should discuss how AI relies on broad datasets that may contain conflicting or outdated information, which leads to contradictions. Students should understand that fact-checking ensures accuracy by verifying AI-generated information against trusted sources. They should also recognize that refining prompts (e.g. specifying a reliable source, adding context, etc) helps AI produce clearer and more relevant responses.

**AI**

Learns from broad, generalized datasets.
Data might be conflicting or outdated.

**Medical Professional**

Deep knowledge on specific subjects
Current methods and research

■ Say: AI models do not have deep expertise like medical professionals. AI relies on broad datasets that may contain conflicting or outdated information, which leads to contradictions. Fact-checking ensures accuracy by verifying AI-generated information against trusted sources. Refining prompts by doing things like specifying a reliable source or adding context helps AI produce clearer and more relevant responses.

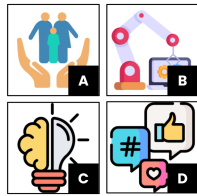
■ Do This: Explain the differences between where an AI and a medical professional acquire information.

AI Contradictions in Practice

Lesson 5

Do This:

1. Go to Lesson 5, Level 3.
2. Practice with the various *forms of questioning* designed to produce contradictions.
3. Add your reflections to the Activity Guide.



AI Contradictions in Practice (10 mins)

■ Transition: Direct students to Lesson 5, Level 3 to complete Levels 3 and 4. Students choose a different topic in this choice level (e.g., government, technology, education, social media). In pairs, students practice with the various forms of questioning provided and discuss the results. Have them add their reflections to the last page of their Activity Guide.

Refining the Prompt

- **Opposite Framing:** Ask "Why is it good to..." vs "Why is it bad to..."
- **Same Framing:** Ask the same question with consistent phrasing.
- **Same Framing + Reliable Source:** Ask the same question and provide a credible source for additional context.



■■ Do This: Click through the animated slide to review the three different ways in which students were asked to refine the prompt in the previous activity.

What is Context?

Context is additional information or data surrounding a prompt or query that helps the AI generate more accurate and relevant responses.



■■ Do This: Define context.

Try Your Own

Lesson 5

Do This:

1. Go to Lesson 5, Level 5.
2. Try writing your own different framings of a question to see how the AI responses differ.



■ Transition: Direct students to Lesson 5, Level 5. In pairs, students practice prompting the AI Chat with different framings of their own questions to see if they can produce AI contradictions and can experience themselves how their framing can alter the response an AI outputs.

Teaching Tip

If students aren't sure what topic to choose, direct them to choose one of the examples from the level:

Mandatory physical education in school

Free higher education

Use of AI in school assignments

The current US voting age

If students need help with prompting the AI, direct to them to modify the prompt examples from the level:

Positive framing

Negative framing

Positive framing with a reliable source

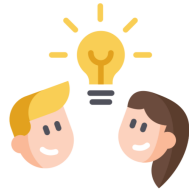
Check Your Understanding

Lesson 5

☐ ☐ ☐ **7** ☐

Do This:

1. Go to Lesson 5, Level 7.
2. Respond to the questions on Levels 7 and 8.



■ ■ Do This: Direct students to Lesson 5, Level 7 to complete the formative assessment on Levels 7 and 8.

Assessment Opportunity

This level can be used as a formative assessment to check student understanding of the lesson objectives. Here are things to look for and actions you can take:

Review responses to see whether students can explain why AI might contradict itself and how that differs from human contradictions. If students struggle to articulate these differences, have them revisit the telehealth example (or another real-world scenario) and prompt them to highlight exactly where the AI's contradiction arose—and contrast it with how a human might revise a diagnosis after new data.

See whether students can identify specific potential harms, and if not, direct them back to the telehealth chatbot examples provided in the lesson. You could also show short clips or articles about AI's role in other high-stakes fields (finance, insurance) to further underscore the importance of verifying AI outputs.

Wrap Up

Wrap Up (5 mins)

Today, you learned about . . .

- Experimenting with AI to uncover contradictions and explain why they occur.
- How AI's reliance on probability-based models can lead to inconsistencies.
- The implications of AI-generated contradictions in real-world applications.

■ ■ Do This: Review the lesson objectives.

Question of the Day

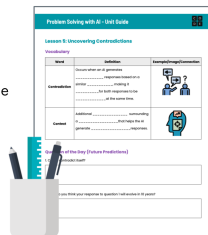
Can AI contradict itself?

■ ■ Do This: Review the Question of the Day.

Future Predictions

You should have:

- Problem Solving with AI Unit Guide
- pen or pencil



Future Predictions (5 mins)

■ ■ Do This: Have students take out their Problem Solving with AI Unit Guide.

✓ **Do This:**

- Answer the Question of the Day in your Unit Guide.
- Predict how the ability of AI to contradict itself might evolve over the next 10 years.

Unit Guide

■ ■ Do This: In their Unit Guide, students answer the Question of the Day.

■ ■ Do This: Students make a prediction about how the ability of AI to contradict itself might evolve over the next 10 years.

✎ **Key Vocabulary**

- **contradiction:** when an AI generates conflicting or opposing responses based on a similar input, making it impossible for both responses to be correct at the same time
- **context:** additional information or data surrounding a prompt or query that helps the AI generate more accurate and relevant responses

■ ■ Do This: Review the Key Vocabulary.